LESSON PLAN · PYTHON (GR 6–8) BEGINNER: PYTHON GAME CODERS

Lesson 2: Variables and Input

Module 1 | 60 minutes | variables, input, f-strings | Grades 6–10 · ages 11–15

Objective: Store information in variables, ask the user questions with input(), and build sentences with f-strings.

Success criteria

- I can store a value in a variable and use it again.
- I can ask the user a question with input().
- I can use an f-string to put variables into a sentence.
- I can choose good variable names.

Key vocabulary

- **variable**: A named box that stores a value.
- **assign**: Put a value into a variable with =.
- **input()**: Asks the user a question and waits for an answer.
- **f-string**: A string with f in front, where {name} is replaced by the variable's value.
- **value**: The information stored in a variable, like "Meera" or 10.

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: friendly_greeter.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Box Swap**	Play Box Swap (unplugged) with three volunteers as variables. The class fills in a silly sentence from the boxes.
5 min	**Hook and video**	Show the hook and ask: "What will print?" Chat poll: type your guess. Play the video with captions; pause at "Spot the bug".
10 min	**Live demo and build-along**	Run the live demo, then everyone types the friendly greeter. Thumbs check: "Did your greeter say your name?"
8 min	**Challenge, predict and sort**	Greeting bot challenge, then predict-the-output and the variable-name sort. Quick competition: fastest correct sort wins.
15 min	**Project in breakout pairs**	In breakout pairs, each learner builds a Mad Libs story, then swaps and plays their partner's game using the chat.
6 min	**Debug race**	Race to fix "the confused greeter". Ask the winner to explain both bugs out loud.

7 min	**Show-and-tell**	Two or three learners run their Mad Libs live while the class shouts out the words. Vote for the funniest story with emoji reactions.
4 min	**Wrap-up and exit ticket**	Fist-to-five on the success criteria. Exit ticket: favourite colour program.

Unplugged warm-up

Box Swap (5 min game): three learners each “become” a variable and hold up a name card (snack, city, hero). The teacher calls out lines like snack = "jalebi" and the learner writes the new value on their whiteboard, rubbing out the old one. Then the teacher reads an f-string sentence aloud and the class fills it in from the boxes.

Teaching the idea

A variable is a named box that stores a value; = means “store”, not “equals”. input() pauses the program, shows a question and returns what the user typed, always as text. f-strings let us drop variables into sentences: put f before the quotes and the variable in curly brackets. Today we use these three ideas to make programs that react to the person using them.

Analogy: Variables are like labelled tiffin boxes: the label (name) stays the same, but you can change what is inside, and you only ever see the latest lunch.

Live demo: 1) Type team = "Mumbai Waves" and on the next line print(team). Run it: point out there are no quotes around team in print. 2) Change it to print("team"), run, and ask what changed (it prints the word team). 3) Type name = input("What is your name? ") and run; type a learner’s first name from the class (first names only). 4) Add print(f"Hi {name}!"), run, then delete the f and run again to show Hi {name}!. Put the f back. 5) Type snack = "samosa" then snack = "vada pav" and print(snack), and ask the class to predict before running.

Questions to ask

Ask	Listen for
What does = do in Python?	It stores the value on the right in the variable on the left.
What type of value does input() give back?	Always text (a string), even if the user types a number.
Why do we need the f in f"Hi {name}"?	Without it, Python prints {name} as plain text.
Is 2total a good variable name?	No, names cannot start with a number.
If snack changes twice, which value is printed?	The latest one.

Differentiation

Support
Give a fill-in template: ___ = input("___ ") and print(f"... {___} ..."). Learners only choose names and questions.

Stretch
Add a second story ending, and use .upper() to SHOUT one word, like {animal.upper()}.

Common misconceptions

Misconception	How to fix it
= means “is equal to” like in maths	In Python, = means “store this value in the box”.
print("name") prints the variable	Quotes make it the word name. Use print(name) or an f-string.
Variable names are not case-sensitive	Age and age are two different variables.
input() gives a number if you type a number	input() always gives text. Next lesson we convert it with int().

Assessment

Use the Python Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write two lines: one that asks for a favourite colour, and one that prints “Your favourite colour is ...” with an f-string.

Homework
Play your Mad Libs game with two people at home. Add one more question and one more story line based on their ideas.

Answer key — Kitabash activities

1. Python Lab: greeting bot (perform · python lab)

Goals checked: Asks two questions with input() · Uses an f-string · Prints “Hi Zoya from Pune!” · Works for anyone

One solution:

```
name = input("What is your name? ")
city = input("Which city? ")
print(f"Hi {name} from {city}!")
```

2. Predict the output: changing boxes (perform · code output)

3. Will Python accept this variable name? (perform · categorize)

Works in Python: player_name, total2, score
Breaks Python: 2total, my name, first-name

4. Project: Mad Libs story generator (perform · python lab)

Goals checked: Asks questions with input() · Runs with 5 answers · Prints a story of 4+ lines

Example program:

```
# Mad Libs story generator
# Made by: Arjun

print("=== MAD LIBS: THE MONSOON MATCH ===")
hero = input("Name of a friend: ")
animal = input("An animal: ")
colour = input("A colour: ")
food = input("A food: ")
action = input("An action word (like dance): ")

print()
print("Here is your story!")
print(f"It was raining hard when {hero} walked onto the cricket ground.")
print(f"Suddenly a {colour} {animal} ran onto the pitch!")
print(f"The {animal} grabbed the ball and started to {action}.")
print(f"{hero} offered it some {food}, and it gave the ball back.")
print(f"Everyone agreed: the {colour} {animal} was the player of the match!")
```

5. Debug it: the confused greeter (probe · python lab)

Goals checked: Says hello to Riya · Says the age · Uses f-strings

One solution:

```
name = input("Your name? ")
age = input("Your age? ")
print(f"Hello {name}!")
print(f"You are {age} years old.")
```

Answer key — worksheet

1. **Key words:** variable = A named box that stores a value. · assign = Put a value into a variable with =. · input() = Asks the user a question and waits for an answer. · f-string = A string with f in front, where {name} is replaced by the variable's value.
2. **Plan your program:** Goals: Asks two questions with input(); Uses an f-string; Prints "Hi Zoya from Pune!"; Works for anyone. One solution: name = input("What is your name? ") city = input("Which city? ") print(f"Hi {name} from {city}!")
3. **Predict the output: changing boxes:** A. I want vada pav!
4. **Will Python accept this variable name?:** Works in Python: player_name, total2, score · Breaks Python: 2total, my name, first-name
5. **Plan your program:** Goals: Says hello to Riya; Says the age; Uses f-strings. One solution: name = input("Your name? ") age = input("Your age? ") print(f"Hello {name}!") print(f"You are {age} years old.")

Kitabash original curriculum.